

Vision-Language Models

Image-Text Alignment, BLIP, Flamingo, LLaVA, and Multimodal LLMs

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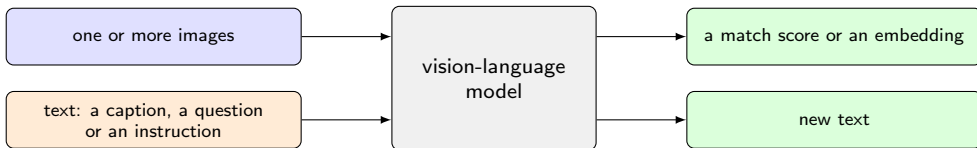
1. Recap and motivation: what vision-language models do, their three families, and the language model side
2. Image-text alignment and joint embedding spaces: contrastive and sigmoid losses, the modality gap, and compositionality
3. BLIP and BLIP-2: one model for matching and captioning, CapFilt, and the Q-Former bridge to a frozen LLM
4. Flamingo: the Perceiver Resampler, gated cross-attention, and few-shot prompting with interleaved images
5. LLaVA: visual instruction tuning, the projector, two-stage training, and LLaVA-1.5
6. Extending LLMs with vision: where vision enters, connectors, and visual-token budgets
7. Training multimodal LLMs: training stages and preference tuning, native resolution, early fusion, grounding, and open models of 2024–2025
8. Evaluation and failure modes: benchmarks, object hallucination, CLIP-blind pairs, and language priors
9. Limitations, summary, and references

- ▶ **Covered earlier:** CLIP trains an image encoder and a text encoder with a symmetric contrastive loss, so matching images and captions land close together in one joint embedding space; zero-shot classification and Stable Diffusion's text encoder both rely on it.
- ▶ **Also covered earlier:** captioning with a CNN encoder and a recurrent or Transformer decoder, attention over image regions, and the BLEU, METEOR and CIDEr scores; cross-attention as the way a diffusion model reads its prompt.
- ▶ **And:** V-JEPA 2's video encoder connected to a large language model through a small projector and then tuned end-to-end with it, and in-context learning as a way to use a foundation model without changing its weights.
- ▶ **Today:** models that read images and write language: a closer look at the joint embedding space, BLIP, three designs that connect vision to a large language model (BLIP-2, Flamingo, LLaVA), the multimodal LLM recipe that generalizes them, and how these models are evaluated and where they fail.

By the end of this lecture you should be able to:

- ▶ **Classify** a vision-language model as a dual encoder, a fusion model or an LLM-based model, and state what each family can and cannot do.
- ▶ **Explain** how the softmax (CLIP) and sigmoid (SigLIP) image-text losses shape a joint embedding space, and **identify** its blind spots: the modality gap and bag-of-words behavior.
- ▶ **Describe** how BLIP trains one network for retrieval, matching and captioning (ITC, ITM, LM, CapFilt), and how the BLIP-2 Q-Former feeds a frozen LLM.
- ▶ **Explain** how Flamingo's Perceiver Resampler, gated cross-attention and per-image masking let a frozen language model read interleaved images, and why interleaved web data gives it few-shot learning.
- ▶ **Describe** visual instruction tuning in LLaVA: generated instruction data, the projector, the two training stages, and the LLaVA-1.5 changes.
- ▶ **Compare** the ways to extend an LLM with vision (entry point, connector, resolution, training stages up to preference tuning, early fusion, boxes as text) and **compute** the visual-token cost of a design.
- ▶ **Choose** benchmarks for a multimodal LLM and **diagnose** its typical failures: object hallucination, CLIP-blind pairs, and answers driven by language priors.

- ▶ A **vision-language model (VLM)** takes one or more images together with text and returns either a **match score or embedding**, as CLIP does, or **new text**, as the LLaVA-style assistant built on a large language model (LLM) does. CLIP was covered earlier; the LLaVA-style assistant is built up in this lecture.



Output: a score or an embedding

- ▶ **Retrieval:** find the caption for an image, or the image for a caption.
- ▶ **Zero-shot classification:** rank class-name prompts by match score.

Output: text

- ▶ **Captioning:** describe the image in a sentence.
- ▶ **Visual question answering (VQA):** answer questions.
- ▶ **Document understanding:** read text, tables and charts.
- ▶ **Grounding:** give a region's box as text, e.g. $(x_1, y_1), (x_2, y_2)$.
- ▶ **Visual chat:** follow multi-turn instructions about images.

Box format (corners scaled to $[0, 1000]$): Bai et al., "Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond," 2023, [arXiv:2308.12966v3](https://arxiv.org/abs/2308.12966v3), Sec. 2.2.

Task	Benchmark	Metric	How one prediction is scored
Retrieval	Flickr30K (1K test), COCO (5K test)	Recall@K	1 if a paired caption (image query) or the paired image (caption query) is in the top K
Zero-shot classification	ImageNet	top-1 accuracy	1 if the best-matching class prompt names the true class
Captioning	COCO (5K test)	CIDEr	consensus with the 5 human captions
VQA	VQAv2	VQA accuracy	$\min(n/3, 1)$ for n matching annotators, averaged over the 10 subsets of 9 annotators
Document VQA	DocVQA (50K questions, 12,767 images)	ANLS	best $s = 1 - \text{NL}(a_j, o)$ over the reference answers a_j , with $s = 0$ when $\text{NL} \geq 0.5$

Recall@K, top-1 accuracy and CIDEr were covered earlier; Flickr30K has 31K images with 5 captions each. **VQA accuracy:** after averaging, an answer given by 1, 2, 3 or ≥ 4 of the 10 annotators scores 0.3, 0.6, 0.9 or 1. **ANLS** (average normalized Levenshtein similarity) averages the best s over all questions; o is the model's answer and NL the edit distance (fewest character insertions, deletions and substitutions) scaled to $[0, 1]$, so a few misread characters keep most of the credit.

Karpathy and Fei-Fei, "Deep Visual-Semantic Alignments for Generating Image Descriptions," CVPR 2015, [OpenAccess](#), Sec. 4. Antol et al., "VQA: Visual Question Answering," ICCV 2015, [OpenAccess](#), Sec. 3 and footnote 2; scores from the official code, visualqa.org/evaluation.html. Biten et al., "Scene Text Visual Question Answering," ICCV 2019, [arXiv:1905.13648v2](#), Sec. 3.4. Mathew et al., "DocVQA: A Dataset for VQA on Document Images," WACV 2021, [arXiv:2007.00398v3](#), Secs. 1 and 5.1.

Three Ways to Combine Vision and Language

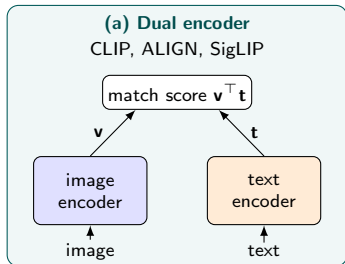
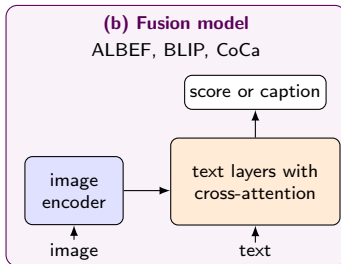
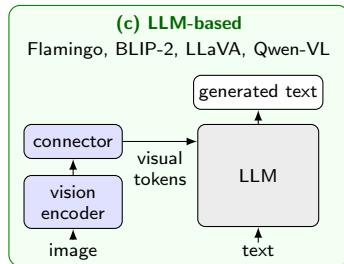


Image embeddings are computed once and indexed, so a text query costs one encoder pass plus dot products. Image and text meet only in that product; no text output.



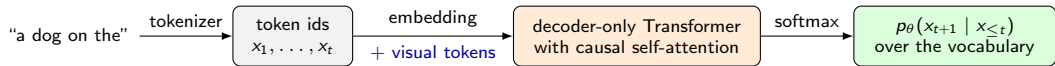
Text layers cross-attend to image features; BLIP and CoCa also write captions. One joint pass per pair, so ALBEF re-ranks only the top k found by dot products.



Open-ended text with the LLM's knowledge and instruction following. Images enter as extra input tokens or via added cross-attention layers (Flamingo); each adds LLM compute.

Left to right, image and text interact more deeply and each image-text pair costs more to process.

Top- k re-ranking: Li et al., "Align before Fuse: Vision and Language Representation Learning with Momentum Distillation," NeurIPS 2021, [arXiv:2107.07651v2](https://arxiv.org/abs/2107.07651v2), Sec. 5.



- **Tokenizer:** cuts text into subword tokens from a fixed vocabulary (Qwen2: 151,643 regular tokens); an embedding table maps each token id to a vector. LLaVA-style models add visual tokens here.
- **Pretraining:** next-token prediction on large text corpora, the autoregressive likelihood (covered earlier); the causal mask lets position t see only $x_{\leq t}$:

$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_\theta(x_t | x_{<t})$$

x_t : the t -th token; $x_{<t}$: the tokens before it; T : sequence length; θ : the model's weights.

- **Instruction tuning:** "finetuning language models on a collection of datasets described via instructions" (FLAN), which "substantially improves zero-shot performance on unseen tasks". Chat models such as Vicuna compute the loss on the response tokens only; the instruction serves as context.

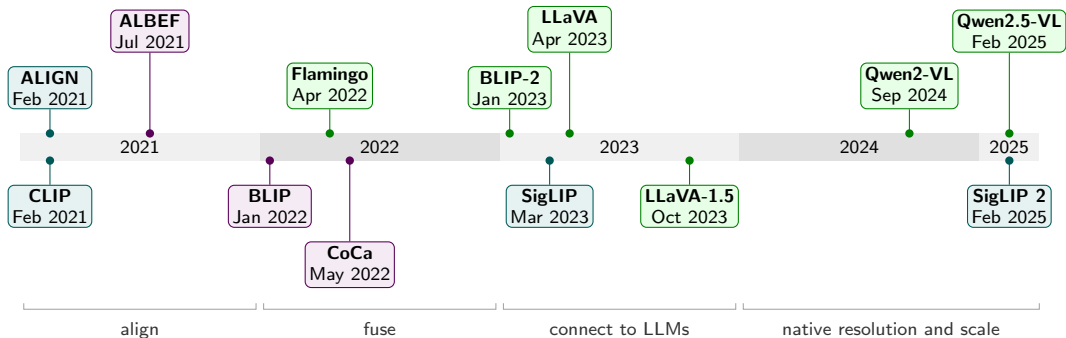
Wei et al., "Finetuned Language Models Are Zero-Shot Learners," ICLR 2022, [arXiv:2109.01652v5](#), Abstract. The Vicuna Team, "Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* ChatGPT Quality," LMSYS blog, 30 Mar. 2023, [lmsys.org/blog/2023-03-30-vicuna](#). Yang et al., "Qwen2 Technical Report," 2024, [arXiv:2407.10671v4](#), Sec. 2.1.

LLM	From	What it is	Used by	Source
Chinchilla 1.4B, 7B, 70B	DeepMind, 2022	decoder-only; the 70B model was trained on 1.4T tokens	Flamingo (frozen)	Hoffmann et al., arXiv:2203.15556v1
OPT 2.7B, 6.7B	Meta AI, 2022	decoder-only; open suite from 125M to 175B	BLIP-2 (frozen)	Zhang et al., arXiv:2205.01068v4
FlanT5 XL, XXL (3B, 11B)	Google, 2022	encoder-decoder T5, instruction-tuned on 1,836 tasks	BLIP-2 (frozen)	Chung et al., arXiv:2210.11416v5
Vicuna 7B, 13B	LMSYS, Mar. 2023	LLaMA (Meta AI) fine-tuned on about 70K ChatGPT conversations shared by users	LLaVA	LMSYS blog; Touvron et al., arXiv:2302.13971v1
Qwen2	Alibaba, 2024	decoder-only; open weights, 0.5B to 72B	Qwen2-VL	Yang et al., arXiv:2407.10671v4
Qwen2.5	Alibaba, 2024	decoder-only; 0.5B to 72B, 18T pretraining tokens	Qwen2.5-VL	Qwen Team, arXiv:2412.15115v2

Frozen: the LLM's weights are not updated when the VLM is trained; given no image, Flamingo falls back to plain language-model behavior.

Which LLM each VLM uses: Alayrac et al., [arXiv:2204.14198v2](#), Sec. 2.2 and App. D.2.2; Li et al., [arXiv:2301.12597v3](#), Sec. 3.4; Liu et al., [arXiv:2304.08485v2](#), Sec. 4.1; Wang et al., [arXiv:2409.12191v2](#), Sec. 2.1; Bai et al., [arXiv:2502.13923v1](#), Sec. 2.1.

A Short History, 2021–2025



Box color gives the family: **dual encoder**, **fusion model**, **LLM-based**. The brackets mark rough eras; the colors show that the families overlap in time. Each box sits at the month of the paper's first arXiv version (v1); full references are at the end.

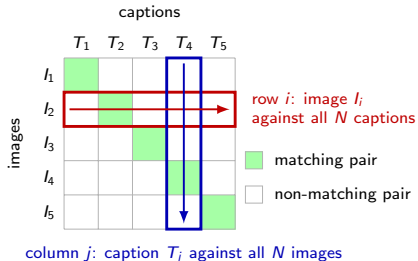
Contrastive Image-Text Alignment (Recap)

- CLIP's symmetric loss (covered earlier), for a batch of N image-text pairs:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}} + \log \frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ji}/\tau}} \right]$$

s_{ij} : cosine similarity of image embedding I_i and text embedding T_j ; τ : learned temperature.

- Term 1 is a softmax along each row (one image against every caption), term 2 along each column (one caption against every image).
- Each row and column is an N -way softmax: the loss couples all N^2 pairs, and normalizing needs the whole matrix.
- CLIP used $N = 32,768$ (about 1.07×10^9 similarities per step), sharded so that each GPU computed only what its local batch needed.



Green: the N matching pairs on the diagonal. Red: one image scored against all N captions. Blue: one caption scored against all N images.

Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021; Sec. 2.5 of the longer [arXiv:2103.00020v1](https://arxiv.org/abs/2103.00020v1) version.

- ▶ **ALIGN** (Jia et al., ICML 2021): CLIP's recipe on **1.8B** image and alt-text pairs (the text a web page stores with an image).
- ▶ **Minimal filtering**: drop small images, alt-texts shared by over 10 images ("1920x1080"), and texts with rare tokens or under 3 or over 20 words.
- ▶ CLIP's pairs came from an allowlist of frequent visual concepts from English Wikipedia; ALIGN keeps the raw distribution.
- ▶ **Encoders**: EfficientNet-L2 and BERT-Large, from scratch; 16,384 pairs per batch.
- ▶ **Zero-shot**: ImageNet 76.4% (CLIP 76.2%); Flickr30K text-to-image R@1 (match ranked first) 75.7 against CLIP's 68.7.
- ▶ In the authors' words, "the scale of our corpus can make up for its noise".



"motorcycle front wheel"



"thumbnail for version as of 21 57 29 june 2010"



"file frankfurt airport skyline 2017 05 jpg"



"file london barge race 2 jpg"



"moustache seamless wallpaper design"



"st oswalds way and shops"

"Example image-text pairs randomly sampled from the training dataset of ALIGN. One clearly noisy text annotation is marked in *italics*." Jia et al., "Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision," ICML 2021, [arXiv:2102.05918v2](https://arxiv.org/abs/2102.05918v2), Figure 2; Abstract, Secs. 2–5, Tables 1 and 4.

- **SigLIP** (Zhai et al., ICCV 2023; named earlier) makes every cell of the $N \times N$ matrix a yes-or-no question:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \log \sigma(z_{ij} (t s_{ij} + b))$$

s_{ij} as in the recap; $z_{ij} = 1$ if $i = j$, else -1 ;

$\sigma(x) = 1/(1 + e^{-x})$; $t = e^{t'}$: learned scale (the role of $1/\tau$); b : learned bias.

- **No normalization over the batch**: each term uses one similarity, so there are no row or column sums, and no extra pass to find the largest logit, which a stable softmax subtracts first.
- **Initialization** $t' = \log 10$, $b = -10$: nearly all N^2 pairs are negatives, so predictions start near that prior (no match) and the first updates do not overcorrect.
- **Ablation** (base-size model, batch 8K): ImageNet zero-shot 63.0% with $b = -10$, 62.0% without b , 61.7% with $b = 0$, 53.7% with $b = 0$ and $t' = \log 1$.

		captions				
		T_1	T_2	T_3	T_4	T_5
images	I_1	+1	-1	-1	-1	-1
	I_2	-1	+1	-1	-1	-1
	I_3	-1	-1	+1	-1	-1
	I_4	-1	-1	-1	+1	-1
	I_5	-1	-1	-1	-1	+1

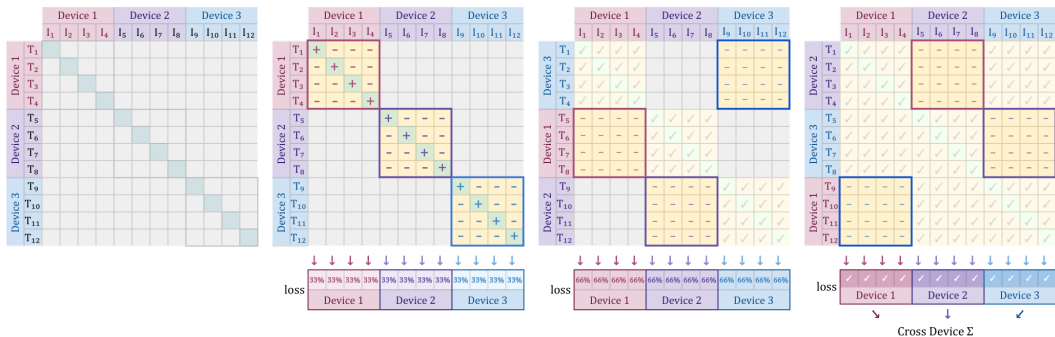
one term:
 $\log \sigma(- (t s_{25} + b))$

$z_{ij} = +1$: N pairs
 $z_{ij} = -1$: $N^2 - N$ pairs

The same batch as in the recap. Each cell is its own logistic-regression example with label z_{ij} ; no term is normalized over its row or column.

Zhai et al., "Sigmoid Loss for Language Image Pre-Training," ICCV 2023, [arXiv:2303.15343v4](https://arxiv.org/abs/2303.15343v4), Sec. 3.2 and Algorithm 1 (form of the loss), Sec. 4.9 and Table 4.

The Sigmoid Loss, Chunked Across Devices



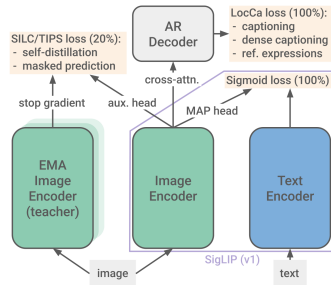
A batch of 12 on 3 devices (columns: images, rows: texts), left to right: start; local 4×4 blocks with the positives; texts move one device on and the new block is added; final sum across devices. Zhai et al., [arXiv:2303.15343v4](https://arxiv.org/abs/2303.15343v4), Figure 1; Sec. 3.3.

- Each device only ever needs its own images and one block of texts at a time, because no term looks at the rest of its row or column.

- ▶ **Memory:** with D devices holding $m = N/D$ pairs each, a device holds one $m \times m$ block at a time and never gathers every embedding: m^2 entries in place of N^2 , enough for batches above one million.
- ▶ **Batch size:** with a locked image tower (SigLiT: a frozen pretrained image encoder, only the text encoder trains), sigmoid beats softmax by a large margin below 16K pairs and both saturate near 32K.
- ▶ **Trained from scratch,** SigLIP peaks at 32K pairs while softmax needs 98K and still does not catch up; a 307K batch hurts both losses.
- ▶ **Cost:** SigLiT reaches 84.5% zero-shot ImageNet on four TPUv4 chips (Google's accelerators) in two days.

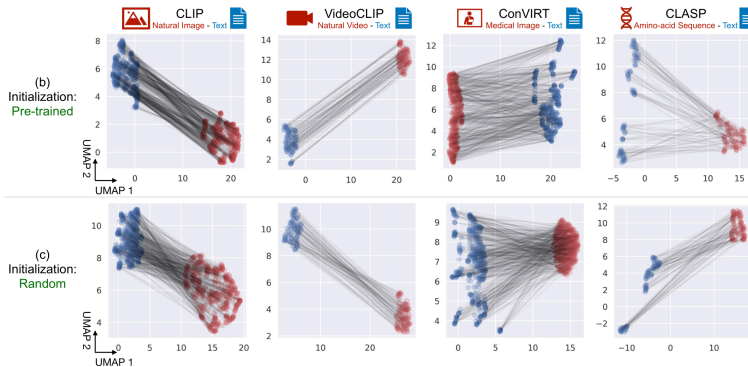
Zhai et al., [arXiv:2303.15343v4](https://arxiv.org/abs/2303.15343v4), Secs. 3.3, 4.1, 4.2 and 4.4, Figure 2 and Table 1.

- ▶ **SigLIP 2** (Tschannen et al., 2025) keeps SigLIP's architecture (swap in its weights and multilingual tokenizer) and trains it with a new recipe on alt-texts in 109 languages.
- ▶ **Decoder** (from LocCa, a location-aware captioner): cross-attends to the image tokens to caption, to locate phrases and to describe boxes; discarded after training.
- ▶ **Last 20%:** self-distillation and masked prediction with an EMA (exponential moving average) teacher, as in DINO and iBOT (covered earlier), for better per-patch features.
- ▶ **NaFlex:** several sequence lengths at each image's native aspect ratio.
- ▶ SigLIP to SigLIP 2, ViT-B/16 (Base-size ViT on 16×16 -pixel patches; L is Large) at 256 pixels: ImageNet zero-shot 76.7 to 79.1%.
- ▶ SigLIP encoders are the vision towers of PaliGemma, LLaVA-OneVision and Gemma 3.



MAP head: attention pooling of the image tokens.
Tschannen et al., "SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features," 2025, [arXiv:2502.14786v1](#), Figure 1; Sec. 2 and Table 1.

Vision towers: Beyer et al., PaliGemma, 2024, [arXiv:2407.07726v2](#), Sec. 3.1; Li et al., LLaVA-OneVision, 2024, [arXiv:2408.03326v3](#), Sec. 3.1; Gemma Team, Gemma 3, 2025, [arXiv:2503.19786v1](#), Sec. 2.1.

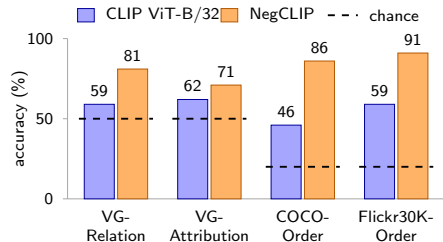


UMAP projections (covered earlier) of four contrastive models (CLIP, VideoCLIP, ConVIRT, CLASP). Top row: pretrained; bottom row: randomly initialized. Red: image, video, medical-image or amino-acid embeddings; blue: text; lines join pairs. Liang et al., "Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning," NeurIPS 2022, [arXiv:2203.02053v2](https://arxiv.org/abs/2203.02053v2), Figure 1(b,c).

- ▶ **Two cones:** image and text embeddings sit in separate regions, in CLIP and in video, medical and protein models (top row of the previous figure).
- ▶ **From initialization:** a deep encoder maps all inputs into a narrow cone (random ResNet: mean cosine 0.99), so two encoders start apart (bottom row).
- ▶ **Kept by training:** at CLIP's $\tau = 1/100$, closing the gap (0.82 between modality centers) raises the loss; fine-tuning at $\tau = 1/10$ or 1 shrinks it.
- ▶ **It matters:** shifting along the gap changed zero-shot accuracy (EuroSAT 54.1 to 56.5%) and cut offensive labels on FairFace faces for all race groups.
- ▶ **In practice:** CLIP ViT-B/32 image-caption cosines mostly lie between about 0 and 0.4; use them to rank, and never compare them with text-text cosines.

Liang et al., [arXiv:2203.02053v2](#), Secs. 2, 4 and 5, Tables 1–2. Cosine range: Hessel et al., “CLIPScore: A Reference-free Evaluation Metric for Image Captioning,” EMNLP 2021, [arXiv:2104.08718v3](#), Sec. 3, footnote 6; CLIPScore (covered earlier) multiplies them by 2.5 for this reason.

- ▶ **ARO** (Attribution, Relation, and Order; Yuksekgonul et al., ICLR 2023), over 50,000 cases: choose the true caption over a swapped one (“the grass is eating the horse”) or over word shuffles.
- ▶ **Why**: retrieval barely needs word order; with caption words shuffled, CLIP still puts a correct caption in its top 5 for 85.4% of Flickr30K images (95.0% intact), so pretraining need not learn order.
- ▶ **NegCLIP**: fine-tune on COCO with word-swapped captions as extra columns and nearest-neighbour images in the batch; CIFAR accuracy falls by at most 1 point, ImageNet from 75 to 72%.
- ▶ **Winoground** (Thrush et al., CVPR 2022): 400 cases of two images and two captions with the same words reordered. Group score (all four comparisons correct): humans 85.5, CLIP ViT-B/32 8.0, chance 16.67.



ARO accuracy (%). VG: Visual Genome images annotated with relations and attributes; Order: the true caption against four word shuffles.

Yuksekgonul et al., “When and why vision-language models behave like bags-of-words, and what to do about it?,” ICLR 2023, [arXiv:2210.01936v3](https://arxiv.org/abs/2210.01936v3), Abstract, Secs. 2 and 4, Tables 3 and 6.

Thrush et al., “Winoground: Probing Vision and Language Models for Visio-Linguistic Compositionality,” CVPR 2022, [arXiv:2204.03162v2](https://arxiv.org/abs/2204.03162v2), Sec. 3 and Table 3.

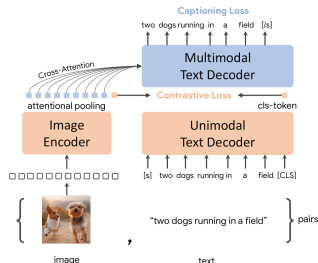
- **CoCa** (Yu et al., 2022) trains an image encoder and a text decoder from scratch on

$$\mathcal{L} = \lambda_{\text{Con}} \mathcal{L}_{\text{Con}} + \lambda_{\text{Cap}} \mathcal{L}_{\text{Cap}}, \quad \lambda_{\text{Con}} = 1, \quad \lambda_{\text{Cap}} = 2$$

$\mathcal{L}_{\text{Con}}$: the recap's loss with $\frac{1}{N}$ in place of $\frac{1}{2N}$;

$\mathcal{L}_{\text{Cap}} = -\sum_t \log p_{\theta}(y_t | y_{<t}, x)$, where p_{θ} is the decoder's probability of caption token y_t given image x and the earlier tokens.

- **Decoder**: the lower half has no cross-attention; its output at an appended [CLS] token (an extra token that summarizes the sequence) is the text embedding for $\mathcal{L}_{\text{Con}}$. The upper half cross-attends to the image and predicts the caption; one forward pass gives both losses.
- **ImageNet**: 86.3% zero-shot, 90.6% with a frozen encoder, 91.0% fine-tuned. In a base-size ablation, adding $\mathcal{L}_{\text{Cap}}$ raised zero-shot accuracy from 70.7 to 71.6%.
- The next model, BLIP, keeps the contrastive and captioning objectives and adds a matching one.



Attentional pooling: learned queries attend over the image tokens (1 query for the contrastive embedding, 256 for the decoder). Yu et al., "CoCa: Contrastive Captioners are Image-Text Foundation Models," 2022, [arXiv:2205.01917v2](https://arxiv.org/abs/2205.01917v2), Figure 2 (left half; pseudocode omitted); Secs. 3.2 and 4, Table 8.

- ▶ **LAION-5B** (named earlier): 5.85B web pairs whose CLIP ViT-B/32 image-text cosine is at least 0.28 (0.26 if not English).
- ▶ **DataComp** (Gadre et al., NeurIPS 2023): fixed training code and compute; entrants compete on the data, mostly by filtering a 12.8B-pair pool (38 test sets).
- ▶ At the 12.8B scale the best filter adds 6.9 ImageNet points over no filtering. **DataComp-1B**: pairs in the top 30% by CLIP ViT-L/14 score whose images also cluster near ImageNet images.
- ▶ **MetaCLIP** (Xu et al., 2023): keep pairs whose text contains one of 500,000 WordNet and Wikipedia entries, at most 20,000 per entry (“photo”: 54M matches, 20K kept).
- ▶ Without that cap, the whole 1.6B matched pool lowers ViT-B/32 from 65.5 to 61.9%.

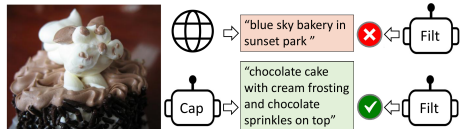
ImageNet zero-shot top-1 (%); within a block the model and training compute match. WIT: CLIP’s own 400M pairs.

Training data	ImageNet
ViT-L/14, 13B samples seen, same compute	
WIT (0.4B)	75.5
LAION-400M (0.4B)	72.8
LAION-2B (2.3B)	73.1
DataComp-1B (1.4B)	79.2
ViT-B/16, CLIP’s training recipe	
WIT (0.4B)	68.3
MetaCLIP (0.4B)	70.8
MetaCLIP (1B)	72.4

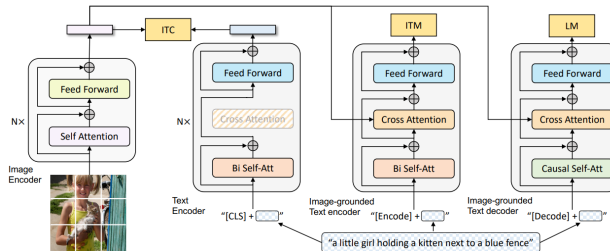
Gadre et al., “DataComp: In search of the next generation of multimodal datasets,” NeurIPS 2023 Datasets and Benchmarks, [arXiv:2304.14108v5](#), Table 1, Secs. 1 and 4; Xu et al., “Demystifying CLIP Data,” 2023, [arXiv:2309.16671v6](#), Sec. 3, Tables 4–6.

Schuhmann et al., “LAION-5B: An open large-scale dataset for training next generation image-text models,” NeurIPS 2022 Datasets and Benchmarks, [arXiv:2210.08402v1](#), Abstract and Sec. 3.1.

- ▶ **BLIP** (Li et al., ICML 2022) addresses two weaknesses of vision-language pretraining at the time.
- ▶ **Model:** encoder-based models (CLIP, covered earlier; ALBEF) are hard to transfer directly to captioning; encoder-decoder models had not been adopted successfully for retrieval.
- ▶ **Data:** gains came largely from scaling up web alt-texts, which “often do not accurately describe the visual content of the images” (Sec. 3.3).
- ▶ **ALBEF** (Li et al., NeurIPS 2021), the predecessor: contrastive alignment of image and text features before cross-attention fusion, plus soft targets from a moving-average copy of the model.
- ▶ **BLIP’s answers:** the multimodal mixture of encoder-decoder (**MED**) and captioning and filtering (**CapFilt**).



“We use a Captioner (Cap) to generate synthetic captions for web images, and a Filter (Filt) to remove noisy captions.” Li et al., “BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation,” ICML 2022, [arXiv:2201.12086v2](https://arxiv.org/abs/2201.12086v2), Figure 1.



ITC: image-text contrastive loss; ITM: image-text matching; LM: language modeling; "same parameters have the same color".
Li et al., [arXiv:2201.12086v2](https://arxiv.org/abs/2201.12086v2), Figure 2.

- ▶ ViT-B/16 or ViT-L/16 image encoder; the text Transformer starts from BERT-base, a Transformer text encoder pretrained by predicting masked words. A task-specific first token, [CLS], [Encode] or [Decode], marks the mode.
- ▶ The grounded encoder adds a cross-attention layer (covered earlier) between self-attention and feed-forward in every block; the decoder makes self-attention causal.
- ▶ **Sharing** all text layers except self-attention: 252M parameters against 361M unshared, at equal or better scores; sharing self-attention too lowers every score (Table 3).

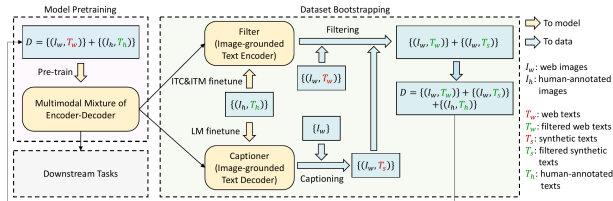
- ▶ **Cost per pair:** one ViT forward pass and three text-Transformer passes, one per mode.
- ▶ **ITC** (unimodal encoders): the contrastive loss (covered earlier) in ALBEF's form. As in MoCo (covered earlier), a momentum encoder, a moving average of the model's weights, fills a queue of extra negatives; it also supplies soft targets q :

$$\mathcal{L}_{\text{itc}} = (1 - \alpha) H(y, p) + \alpha \text{KL}(q \parallel p)$$

p : softmax of the similarities s/τ over the batch and the queue; y : one-hot target; q : the same softmax from momentum features; H : cross-entropy; α : mixing weight (0.4 in ALBEF). Averaged over both directions; q gives partial credit to negatives that also fit.

- ▶ **ITM** (grounded encoder): a linear head on the [Encode] output classifies the pair as matched or not (cross-entropy). **Hard negatives:** each image gets one wrong in-batch caption, drawn with a probability that rises with its ITC similarity, and each caption one wrong image.
- ▶ **LM** (decoder): $\mathcal{L}_{\text{lm}} = -\sum_{t=1}^L \log p_{\theta}(w_t \mid w_{<t}, I)$ over the caption tokens $w_1, \dots, w_L$ of image I , with label smoothing 0.1 (10% of each target spread over the vocabulary). It replaces ALBEF's masked language modeling, so the model can write captions.

Li et al., "Align before Fuse: Vision and Language Representation Learning with Momentum Distillation," NeurIPS 2021, [arXiv:2107.07651v2](#), Section 3; Li et al., [arXiv:2201.12086v2](#), Section 3.2.



I : images, T : texts; subscripts w : web, s : synthetic, h : human-annotated; red: unfiltered, green: filtered or human. Li et al., [arXiv:2201.12086v2](https://arxiv.org/abs/2201.12086v2), Figure 3.

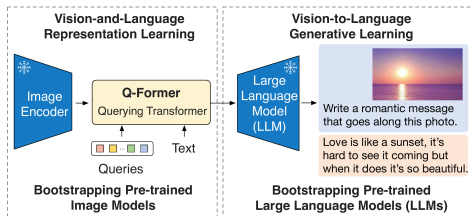
- ▶ Two copies of the pretrained MED are each fine-tuned, lightly, on COCO captions (covered earlier).
- ▶ **Captioner** (grounded decoder, LM): one synthetic caption T_s per web image I_w , by **nucleus sampling**: each token comes from the top tokens whose probabilities sum to $p = 0.9$.
- ▶ **Filter** (grounded encoder, ITC and ITM): removes a web text T_w or a synthetic text T_s when its ITM head predicts “unmatched”.
- ▶ The kept pairs plus the human pairs pretrain a **new** model. Data: 14M images (human-annotated COCO and Visual Genome; web caption sets CC3M and CC12M (Conceptual Captions) and SBU Captions) or 129M with 115M LAION images.

Pretraining data	C	F	TR@1	IR@1	B@4	CIDEr	NoCaps CIDEr
14M images	X	X	78.4	60.7	38.0	127.8	102.2
	X	✓ _B	79.1	61.5	38.1	128.2	102.7
	✓ _B	X	79.7	62.0	38.4	128.9	103.4
	✓ _B beam	✓ _B	79.6	61.9	38.4	128.9	103.5
	✓ _B	✓ _B	80.6	63.1	38.6	129.7	105.1
129M images	X	X	79.6	62.0	38.8	130.1	105.4
	✓ _B	✓ _B	81.9	64.3	39.4	131.4	106.3
	✓ _L	✓ _L	81.2	64.1	39.7	133.3	109.6

C / F: captioner / filter on ViT-B (*B*) or ViT-L (*L*); beam: beam search, which aims for the most probable caption, instead of nucleus sampling. Pretrained model: ViT-B/16 in every row. TR@1 / IR@1: Recall@1 (share of queries whose top result is correct) for image-to-text / text-to-image retrieval on COCO after fine-tuning; B@4: BLEU-4 (BLEU and CIDEr covered earlier); NoCaps: zero-shot captioning by the COCO-fine-tuned model. Li et al., [arXiv:2201.12086v2](https://arxiv.org/abs/2201.12086v2), Tables 1, 2 and 4.

- ▶ Each module helps alone; together they add 2.2 TR@1 and 2.9 NoCaps CIDEr at 14M. The gain holds at 129M, and ViT-L modules lift NoCaps CIDEr to 109.6.
- ▶ **Diversity:** nucleus sampling beats beam search though the filter rejects more of its captions (25% against 19%); beam search “tends to generate safe captions that are common in the dataset” (Sec. 4.3).
- ▶ Fine-tune the two modules separately: with shared weights the filter rejects only 8% of the synthetic captions and every score drops (Table 4).

- ▶ **BLIP-2** (Li et al., ICML 2023) freezes a pretrained image encoder and a pretrained LLM and trains only a bridge between them, which cuts compute and keeps what both models already know.
- ▶ **Q-Former** (Querying Transformer), initialized from BERT-base: 32 learnable queries of width 768 read the frozen image features by cross-attention. Output 32×768 , against 257×1024 ViT-L/14 features.
- ▶ **Frozen encoder:** CLIP ViT-L/14 or EVA-CLIP ViT-g/14 (about 1B parameters). **Frozen LLM:** OPT (decoder-only, no instruction tuning) or FlanT5 (instruction-tuned encoder-decoder).
- ▶ **Two stages:** representation learning with the frozen encoder alone, then generative learning through the frozen LLM.

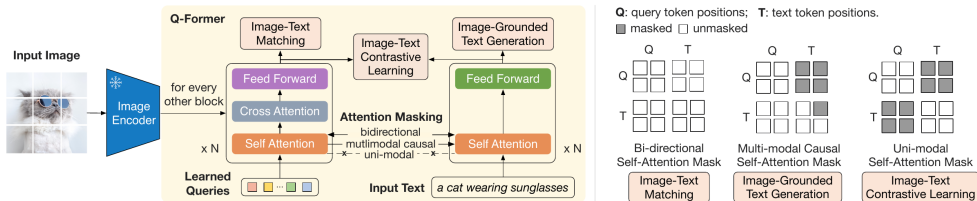


Snowflake: frozen. Li et al., "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models," ICML 2023, [arXiv:2301.12597v3](https://arxiv.org/abs/2301.12597v3), Figure 1.

Largest model (ViT-g with FlanT5-XXL): under 6 days for stage 1 and under 3 days for stage 2 on one machine with 16 A100 (40 GB) GPUs.

Result: zero-shot VQAv2 accuracy (test-dev) of 65.0 against 56.3 for Flamingo-80B, which trains 10.2B of its 80B parameters; BLIP-2 trains 188M, $54\times$ fewer (Table 1; Table 2 lists 108M for this model).

Q-Former: Three Objectives, Three Attention Masks

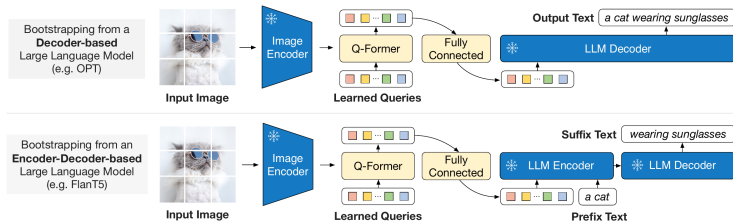


Left: queries and text share the self-attention layers; only the queries cross-attend to the frozen encoder, in every other block.

Right: one mask per objective (gray: masked). Li et al., [arXiv:2301.12597v3](https://arxiv.org/abs/2301.12597v3), Figure 2.

- ▶ **ITC, unimodal mask:** queries and text cannot see each other; score = $\max_k s(\mathbf{h}_k, \mathbf{t})$ over query outputs $\mathbf{h}_k$ and the text [CLS] output $\mathbf{t}$; in-batch negatives replace BLIP's queue, as more samples fit per GPU.
- ▶ **ITG (image-grounded text generation), multimodal causal mask:** queries see only queries; text sees all queries and earlier text, so the queries must carry what the caption needs.
- ▶ **ITM, bidirectional mask:** all positions see each other; a two-class linear head on each output query, logits averaged; hard negatives as in BLIP.
- ▶ ITG also helps retrieval (Table 6): adding it to ITC and ITM in COCO fine-tuning lifts Recall@1 from 84.5 to 85.4 (image to text) and from 67.2 to 68.3 (text to image).

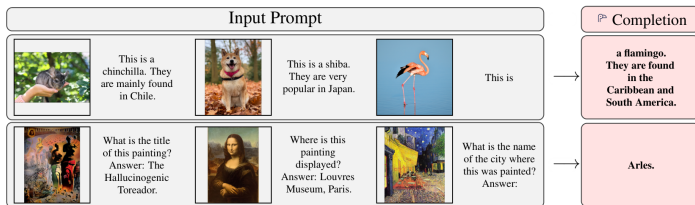
BLIP-2 Stage 2: Soft Visual Prompts for the LLM



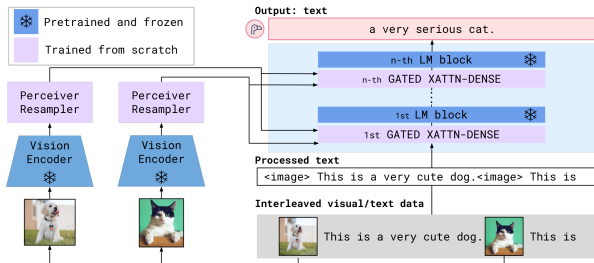
Top: decoder-only LLM (OPT); bottom: encoder-decoder LLM (FlanT5). Li et al., [arXiv:2301.12597v3](https://arxiv.org/abs/2301.12597v3), Figure 3.

- ▶ A linear layer maps the 32 query outputs to the LLM's embedding width; prepended to the text embeddings, they act as **soft visual prompts**. Only the Q-Former and this layer train.
- ▶ **OPT**: language-modeling loss on the text. **FlanT5**: prefix language modeling; the text is split, the prefix enters the encoder with the visual prompts, the decoder generates the suffix.
- ▶ Without stage 1, zero-shot VQA is much worse for both LLMs, and with OPT the accuracy drops sharply as training goes on (Figure 5). In-context examples do not help, which the authors trace to one image-text pair per training sample.
- ▶ **InstructBLIP** (Dai et al., 2023) also feeds the instruction text to the Q-Former, where it meets the queries in self-attention, so the visual features depend on the task.

- **Flamingo** (Alayrac et al., DeepMind, NeurIPS 2022) reads text interleaved with images or videos and writes text; CLIP-style dual encoders (covered earlier) only score image-text pairs.
- **Goal:** in-context learning (covered earlier) with images, as GPT-3, a 175B-parameter text-only language model, does with text. The prompt holds a few (image, text) examples and then the query image; the model completes the text, with no weight update.
- **Answers:** open-ended tasks such as captioning or visual question answering (VQA) use the generated text; multiple-choice tasks rank the candidates by their log-likelihood under the model.



Two few-shot prompts and the completions of Flamingo-80B. Alayrac et al., “Flamingo: a Visual Language Model for Few-Shot Learning,” NeurIPS 2022, [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 1 (first two rows).



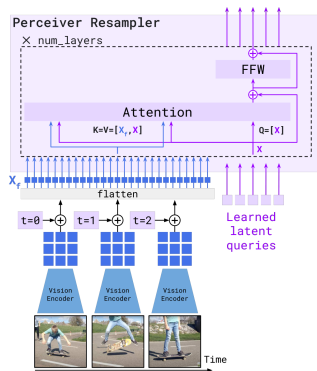
Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 3.

- **Frozen vision encoder:** NFNNet-F6, a ResNet with no normalization layers, which the authors pretrained with CLIP's contrastive loss (covered earlier).
- **Frozen language model (LM):** Chinchilla, DeepMind's text-only LLM.
- **Trained from scratch:** the Perceiver Resampler and the GATED XATTN-DENSE layers (gated cross-attention plus a feed-forward layer) between frozen LM blocks.

Model	Frozen LM	Trainable GATED XATTN-DENSE	Total
Flamingo-3B	1.4B, 24 blocks	1.2B: 24 layers, one before every LM block	3.2B
Flamingo-9B	7.1B, 40 blocks	1.6B: 10 layers, one before every 4th block	9.3B
Flamingo-80B	70B, 80 blocks	10B: 12 layers, one before every 7th block	80B

All three also contain the frozen NFNNet-F6 (435M parameters; Brock et al., ICML 2021) and a trainable Perceiver Resampler (194M); the LMs are Chinchilla models (Hoffmann et al., 2022). Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Tables 4 and 5.

- **Input:** each frame's feature grid from the frozen encoder, plus a learned time embedding per frame, flattened into one sequence X_f (an image is a one-frame video).
- **Latents:** 64 learned queries X attend to keys and values $[X_f, X]$ (features and latents concatenated), then pass a feed-forward layer; 6 such layers.
- **Output:** always 64 visual tokens, whatever the image resolution or the number of video frames, so the LM's cross-attention cost does not grow with either.
- **Origin:** the Perceiver (Jaegle et al., ICML 2021) passes large inputs through a small learned latent array to avoid quadratic attention over the input. BLIP-2's Q-Former (earlier in this lecture) later used 32 such queries.
- **Ablation** (Flamingo-3B, 4 shots; score: each result as a percentage of the state of the art, averaged over five benchmarks): Resampler 70.7; MLP 66.6 and plain Transformer 66.7, both slower.



The figure draws a three-frame video and five output tokens; the model outputs 64. Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 5 (pseudo-code omitted); ablation: Table 3.

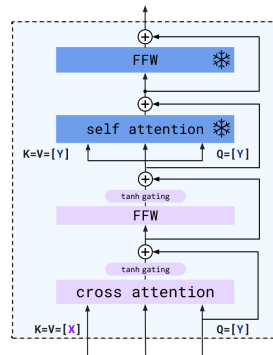
Each GATED XATTN-DENSE block updates the language features y with the visual tokens x , then passes y to the next frozen LM block:

$$y \leftarrow y + \tanh(\alpha_{\text{xattn}}) \cdot \text{Attn}(q=y, kv=x)$$

$$y \leftarrow y + \tanh(\alpha_{\text{dense}}) \cdot \text{FFW}(y)$$

$\alpha_{\text{xattn}}, \alpha_{\text{dense}}$: learnable scalars of the block, initialized to 0; Attn: attention with queries from y , keys and values from x ; FFW: feed-forward layer.

- **At initialization** $\tanh(0) = 0$, so the new branches add nothing and the model computes exactly the pretrained LM; the authors report better training stability and final performance.
- **Ablations** (score 70.7): no tanh gate 66.5, with unstable training; vanilla Transformer-decoder cross-attention 66.9; layers stacked on top of the unchanged LM 63.1.
- **Freezing the LM** prevents catastrophic forgetting of what pretraining taught: fine-tuning the LM scores 62.7, training it from scratch 57.8.



One GATED XATTN-DENSE block (purple, trained) feeding a frozen LM block (blue). Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 4 (middle panel); ablations: Table 3.

- ▶ **One text stream:** a web page or a few-shot prompt becomes text with an `<image>` tag at each image's position. A learned `<EOC>` (end of chunk) token closes the text before each image and at the end; `<BOS>` (beginning of sequence) starts it.
- ▶ **Per-image attention masking:** let $\phi(\ell)$ be the index of the last image before text position ℓ (0 if none). Token ℓ cross-attends only to the visual tokens of image $\phi(\ell)$; all others are masked.

Text chunk	$\phi(\ell)$	Image 1 tokens	Image 2 tokens
<code><BOS>Cute pics of my pets!<EOC></code>	0	masked	masked
<code><image>My puppy sitting in the grass.<EOC></code>	1	visible	masked
<code><image>My cat looking very dignified.<EOC></code>	2	masked	visible

Example and mask after Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 7, with one row per chunk where the paper draws one per token; ϕ from Appendix A.1.3, ablation from Table 10.

- ▶ **Earlier images still count:** the frozen LM's causal self-attention reads every earlier text token, and those tokens have already attended to their own image. Trained on sequences of at most 5 images, the model still benefits from prompts with up to 32 (image, text) shots.
- ▶ **Ablation:** cross-attending to all previous images lowers the score from 70.7 to 63.5. In training, a random half of the web pages map text to the next image instead (App. A.3.2).

Dataset	Content	Size	λ_m
M3W	MultiModal MassiveWeb: web pages with their images kept in place	about 43M pages	1.0
ALIGN	images with their alt-text (earlier in this lecture)	1.8B pairs	0.2
LTIP	Long Text & Image Pairs: 20.5-token captions on average (ALIGN: 12.4)	312M pairs	0.2
VTP	Video & Text Pairs: short videos (about 22s) with one sentence each	27M videos	0.03

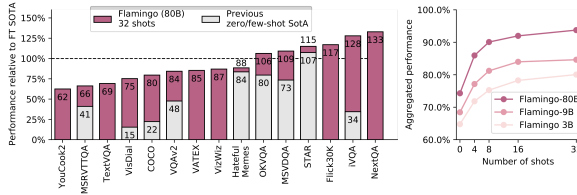
- **Objective:** a weighted sum of per-dataset negative log-likelihoods of the text given the preceding images; the gradients of all $M = 4$ datasets are summed at every step.

$$\sum_{m=1}^M \lambda_m \mathbb{E}_{(x,y) \sim \mathcal{D}_m} \left[- \sum_{\ell=1}^L \log p(y_\ell \mid y_{<\ell}, x_{\leq \ell}) \right]$$

$\mathcal{D}_m$: dataset m , with weight λ_m ; y_ℓ : the ℓ -th of L text tokens; $y_{<\ell}$: the tokens before it; $x_{\leq \ell} = \{x_i : i \leq \phi(\ell)\}$: the images before it.

- **Web data only**, none annotated for machine learning; pairs are written as <image>caption<EOC> to match M3W. The paper ties the few-shot ability to M3W: without it the score falls from 70.7 to 53.4 (without the image-text pairs: 60.9; round-robin updates in place of summed gradients: 62.9).

Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Section 2.4, Appendices A.3.3 and B.1.2, Table 3.



Right: performance rises with shots and model size. Alayrac et al., [arXiv:2204.14198v2](https://arxiv.org/abs/2204.14198v2), Figure 2, Section 1.

- **Results:** one set of weights, 16 benchmarks. With 32 shots Flamingo-80B beats the fine-tuned state of the art on 6 (bars above the dashed line), using about 1000 times less task-specific data.
- **Open versions:** OpenFlamingo (Awadalla et al., 2023), five 3B to 9B models averaging 80–89% of the matching Flamingo on seven datasets; IDEFICS (9B, 80B), trained with OBELICS (141M open web pages, 353M images; Laurençon et al., 2023), is often on par with Flamingo.
- **Still in use:** cross-attention layers inside a frozen LLM, as in Flamingo, are one of the two ways of adding vision compared later in this lecture.

Sources: Awadalla et al., “OpenFlamingo: An Open-Source Framework for Training Large Autoregressive Vision-Language Models,” 2023, [arXiv:2308.01390v2](https://arxiv.org/abs/2308.01390v2); Laurençon et al., “OBELICS: An Open Web-Scale Filtered Dataset of Interleaved Image-Text Documents,” 2023, [arXiv:2306.16527v2](https://arxiv.org/abs/2306.16527v2).

- ▶ **The gap.** Instruction tuning needs instruction-response pairs. Image-caption pairs were plentiful, image instructions with answers scarce, and crowd-sourcing them “time-consuming and less well-defined”.
- ▶ **LLaVA** (Large Language and Vision Assistant; Liu et al., NeurIPS 2023), presented as the first attempt to extend instruction tuning to images, contributes:
 - **data:** 158K image instructions with answers, written by a text-only GPT-4;
 - **a model:** CLIP encoder, one projection matrix, Vicuna; fine-tuned on ScienceQA (21K multiple-choice science questions) and ensembled with GPT-4, a new state of the art there: 92.53%;
 - **a benchmark:** LLaVA-Bench, scored by a GPT-4 judge;
 - open data, code, checkpoints and demo.
- ▶ This is the LLaVA-style recipe (covered earlier): frozen encoder, small projector, instruction tuning.

Prompt (an example from the GPT-4 report; the photo shows a man ironing clothes on a board fixed to the back of a yellow taxi): “What is unusual about this image?”

Model	Answer (verbatim)
BLIP-2	a man is sitting on the back of a yellow cab
OpenFlamingo	The man is drying his clothes on the hood of his car.
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity. . .

BLIP-2 and OpenFlamingo (both earlier in this lecture) describe the scene; LLaVA answers the question. Liu et al., “Visual Instruction Tuning,” NeurIPS 2023, [arXiv:2304.08485v2](https://arxiv.org/abs/2304.08485v2), Table 3.

- ▶ **Cheap baseline:** make the caption $\mathbf{X}_c$ of an image $\mathbf{X}_v$ the answer to a request $\mathbf{X}_q$ such as “Describe the image concisely.” It “lacks diversity and in-depth reasoning”.
- ▶ **Symbolic image:** the teacher is language-only GPT-4, so each COCO image (covered earlier) is written out as text: its five captions and, for descriptions and reasoning, its object boxes (class name, four coordinates).
- ▶ **Prompt** (conversation type): “You are an AI visual assistant, and you are seeing a single image.” A few hand-written examples per type serve as in-context examples (covered earlier), the only human annotations in the pipeline.
- ▶ In early tests GPT-4 wrote better data than ChatGPT, for example on spatial reasoning.

Response type	What GPT-4 is asked to write	Samples
Conversation	multi-turn questions and answers on object types, counts, actions, locations and relative positions; only questions with definite answers	58K
Detailed description	a full description, requested by a question drawn at random from a fixed list	23K
Complex reasoning	questions whose answers need step-by-step reasoning about the scene	77K
Total	LLaVA-Instruct-158K, on 80K unique COCO images	158K

Liu et al., [arXiv:2304.08485v2](https://arxiv.org/abs/2304.08485v2), Sections 3 and 5.1; prompt from Table 13.

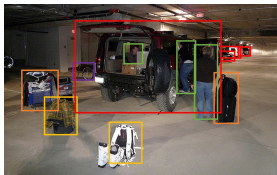


Photo and text adapted from Liu et al.,
[arXiv:2304.08485v2](#), Table 1 (full text in Table 14;
cuts marked ...). GPT-4 never saw the photo.

What GPT-4 reads

Captions (2 of 5): A group of people standing outside of a black vehicle with various luggage. People try to fit all of their luggage in an SUV.

Boxes (2 of 12; not given for conversations): person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], ...

GPT-4 writes as if it saw the photo; the authors later note these descriptions “may contain a small amount of hallucinated content” (Liu et al., “Improved Baselines with Visual Instruction Tuning,” CVPR 2024, [arXiv:2310.03744v2](#), Sec. 5.2).

What GPT-4 writes

1. Conversation (three turns; the first shown)

Q: What type of vehicle is featured in the image?

A: The image features a black sport utility vehicle (SUV).

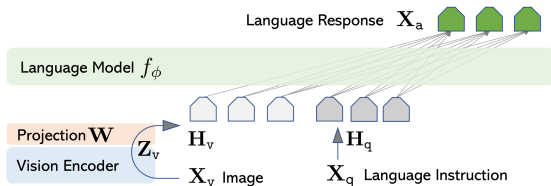
2. Detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, ... They are all working together to pack their luggage into the SUV for a trip. ...

3. Complex reasoning

Q: What challenges do these people face?

A: ... They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...



Liu et al., [arXiv:2304.08485v2](https://arxiv.org/abs/2304.08485v2), Figure 1.

$$H_v = W \cdot Z_v, \quad Z_v = g(X_v)$$

X_v : the image; g : the frozen CLIP image encoder; Z_v : its grid of patch features; W : a trainable projection matrix; H_v : visual tokens as wide as the LLM's word embeddings; H_q : the embedded instruction X_q ; f_ϕ : the LLM (Vicuna) with weights ϕ ; X_a : the answer.

Widths and layer: released LLaVA-13B config, huggingface.co/liuhaotian/LLaVA-13b-delta-v0. Layer ablation: Liu et al., [arXiv:2304.08485v2](https://arxiv.org/abs/2304.08485v2), Section 5.2 and Table 8.

- **Encoder:** CLIP ViT-L/14 (covered earlier), the Large ViT on 14×14 -pixel patches. LLaVA reads the grid features from before its last layer; in an ablation they beat the last layer's, which the authors attribute to more local detail.
- **Projection:** one linear map, shared by all grid features, from 1,024 to 5,120 dimensions for the 13B LLM: about 5.2M weights.
- **Tokens:** a 224×224 image gives $(224/14)^2 = 16 \times 16 = 256$ visual tokens, placed before or after the first question.
- **Why so simple:** a light connector lets data experiments iterate fast; Flamingo's gated cross-attention and BLIP-2's Q-Former were left for future work.

Both stages maximize the likelihood of the assistant's answers:

$$p(\mathbf{X}_a \mid \mathbf{X}_v, \mathbf{X}_q) = \prod_{i=1}^L p_{\theta}(x_i \mid \mathbf{X}_v, \mathbf{X}_{q,<i}, \mathbf{X}_{a,<i})$$

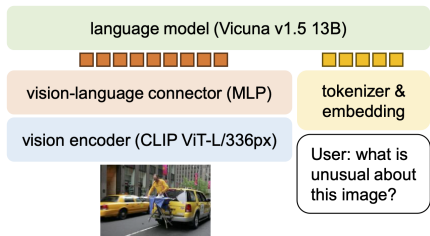
x_i : the token being predicted; $\mathbf{X}_{q,<i}$, $\mathbf{X}_{a,<i}$: the instruction and answer tokens before it; L : sequence length; θ : the trainable parameters. Only answer tokens and the ### stop marker carry loss.

	Stage 1: feature alignment	Stage 2: end-to-end fine-tuning
Data	595K CC3M pairs, balanced over concepts; the caption answers a brief-description request	LLaVA-Instruct-158K, the three response types sampled uniformly
Trained	$\theta = \mathbf{W}$; CLIP and LLM frozen	$\theta = \{\mathbf{W}, \phi\}$; CLIP frozen
Cost	1 epoch, within 4 hours on 8 A100 GPUs	3 epochs, within 10 hours on 8 A100 GPUs

- **Stage 1** trains “a compatible visual tokenizer for the frozen LLM”; in the variant fine-tuned on ScienceQA in stage 2, skipping it lowers accuracy from 90.92% to 85.81%.
- **Stage 2** lifts the LLaVA-Bench (COCO) relative score from 21.5% to 85.1%: a GPT-4 judge rates LLaVA’s answers against those of a text-only GPT-4 that read the true captions and boxes.

CC3M: Conceptual Captions, about 3.3M web images with cleaned alt-text captions (Sharma et al., ACL 2018, aclanthology.org/P18-1238). Liu et al., [arXiv:2304.08485v2](https://arxiv.org/abs/2304.08485v2), Secs. 4.2 and 5, App. C, Tables 2, 4 and 8.

- ▶ **Connector:** a two-layer MLP replaces the linear **W**, the linear-to-MLP switch that helped SimCLR (covered earlier).
- ▶ **Encoder:** CLIP ViT-L/14 at 336 px, the highest resolution CLIP offers: $(336/14)^2 = 24 \times 24 = 576$ visual tokens.
- ▶ **Format prompt:** LLaVA “tends to answer yes for yes/no questions” and falls short on one-word-answer benchmarks. LLaVA-1.5 appends “Answer the question using a single word or phrase.” to short-answer visual question answering (VQA) data, so the answer format follows the prompt.
- ▶ **Data:** 558K BLIP-captioned pairs for alignment, then 665K instructions: LLaVA-Instruct-158K, 40K text-only ChatGPT conversations (ShareGPT), and eight VQA, region and OCR (text-reading) sets such as VQAv2 and RefCOCO (referring expressions: phrases that each pick out one object in a COCO image); about 1.2M public samples.
- ▶ **Cost:** about 6 hours of alignment and 20 hours of instruction tuning on one 8-A100 node, about a day for the 13B model; state of the art on 11 of its 12 benchmarks.

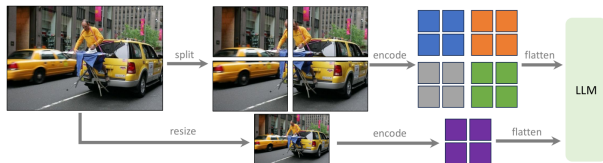


Liu et al., [arXiv:2310.03744v2](https://arxiv.org/abs/2310.03744v2), Figure 1 (right panel); Sections 3.1–3.3 and 4, Tables 3 and 7.

MME-Perception (yes/no questions), 7B models: 809.6 for LLaVA, 1323.8 once VQAv2 and the format prompt are added, 1510.7 for LLaVA-1.5 (Table 2).

558K set: LLaVA repository, github.com/haotian-liu/LLaVA (docs/Data.md).

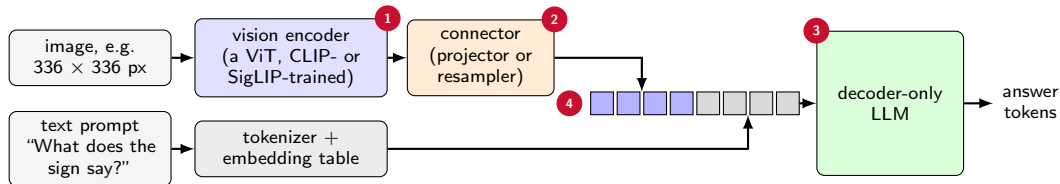
Higher Resolution by Tiling the Image



LLaVA-1.5-HD. Liu et al., [arXiv:2310.03744v2](https://arxiv.org/abs/2310.03744v2), Figure 2, Sec. 3.4, App. A.1; the LLaVA-NeXT post uses the same diagram.

- **LLaVA-1.5-HD:** open CLIP encoders then stopped at 336 px, so HD tiles the padded image at its encoder's native 224 px (up to six tiles, e.g. 2×3 or 1×6) and encodes each tile alone. A 224 px whole-image view is added, and a row-end token closes each feature row.
- **LLaVA-NeXT AnyRes** (Jan. 2024): 336 px tiles; the post lists 2×2 , $1 \times \{2, 3, 4\}$ and $\{2, 3, 4\} \times 1$ grids, up to 1344×336 (released configs stop at 1008×336), “4x more pixels” than LLaVA-1.5.
- **Tokens:** a 2×2 grid of 336 px tiles plus a 336 px global view gives $4 \times 576 + 576 = 2,880$ visual tokens before row-end tokens: $5 \times$ LLaVA-1.5 and over $11 \times$ LLaVA.
- **Context:** Vicuna v1.5 reads 4,096 tokens: one such image fills about 70% of them before any text, and two do not fit.

Source: Liu et al., “LLaVA-NeXT: Improved reasoning, OCR, and world knowledge,” LLaVA blog, 30 Jan. 2024, llava-vl.github.io/blog/2024-01-30-llava-next. Global view: `llava/mm_utils.py`, github.com/haotian-liu/LLaVA; context: [llava-v1.6-vicuna-13b config](#).



Blue: visual tokens; gray: text tokens; red: the four design decisions. The tokenizer splits text into subword units, which the embedding table maps to vectors.

- **Training signal:** predict the next text token, given the image and the text before it:

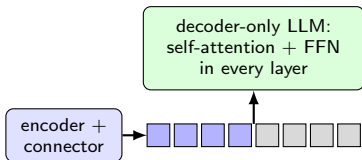
$$\mathcal{L}(\theta) = - \sum_{i \in \mathcal{A}} \log p_{\theta}(x_i | \mathbf{H}_v, x_{<i})$$

$\mathbf{H}_v$: visual tokens, as in LLaVA; $x_{<i}$: earlier text tokens; $\mathcal{A}$: positions that carry loss (the answer tokens in LLaVA's instruction tuning, all text tokens in Flamingo's pretraining); θ : the unfrozen weights, connector included.

1. **Encoder:** which ViT, trained how, at what resolution.
2. **Connector:** projector or learned-query resampler.
3. **Entry point:** input sequence or cross-attention.
4. **Token budget:** visual tokens per image.

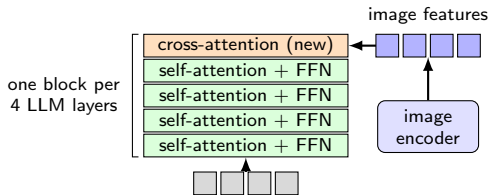
BLIP-2 (Q-Former, input sequence), Flamingo (Perceiver Resampler, cross-attention) and LLaVA (projector, input sequence) are three settings of decisions 2 and 3.

(a) Visual tokens in the input sequence



- **Used by** LLaVA, BLIP-2 (Q-Former outputs prepended as “soft visual prompts”) and Qwen-VL (256 tokens between `<img>` and `</img>`). No new LLM layers; the LLM may stay frozen (BLIP-2) or be tuned (LLaVA).
- **Cost:** every visual token takes a context position and passes through self-attention and the FFN (the per-token MLP of each block, FFW in Flamingo’s figures) in every layer.

(b) Cross-attention layers inside the LLM



- **Used by** Flamingo (every 7th layer in Flamingo-80B) and Meta’s Llama 3 (every 4th, as drawn). Text tokens query the image features: the sequence keeps its length, and image tokens skip the LLM’s FFNs.
- **Cost:** new weights, about 100B in the cross-attention layers of the 405B Llama 3. The LLM can stay frozen; Llama 3 freezes it “to maintain text-only performance”.

Llama 3 feeds speech in by route (a) and vision by route (b). Sources: Llama Team, AI @ Meta, “The Llama 3 Herd of Models,” 2024, [arXiv:2407.21783v3](#), Sec. 7, 7.2, 7.5.2, 8.2.1; BLIP-2, [arXiv:2301.12597v3](#), Sec. 3.3; Qwen-VL, [arXiv:2308.12966v3](#), Sec. 2.1–2.2; Flamingo, [arXiv:2204.14198v2](#), Sec. 3.3.

Connector	Tokens per image	Example	What it trades
Linear projection	256 at 224 px	LLaVA	one per patch; keeps local detail
Two-layer MLP	576 at 336 px	LLaVA-1.5	one per patch; more capacity
Q-Former	32 learned queries	BLIP-2	fixed and short; own pretraining
Perceiver Resampler	64 learned latents	Flamingo	fixed at any resolution or frame count
Cross-attention resampler	256 from 1,024 patches	Qwen-VL	one layer; 2D position encodings limit layout loss
Pixel unshuffle + MLP	256 per 448-px tile	InternVL 1.5	folds 32×32 patch features into 16×16 vectors with $4\times$ the channels (the paper calls it pixel shuffle); count grows with the tiles
2×2 merge + MLP	one per 28×28 px	Qwen2-VL	an MLP merges each 2×2 group; count follows the image area

- **Two properties a connector needs** (Honeybee, Cha et al., CVPR 2024): “flexibility in managing the number of visual tokens”, which learned-query resamplers have, and “preservation of local context from visual features”, which per-patch projectors have. 2×2 folding sits between them: a fixed $4\times$ cut that keeps each block’s neighbours together.

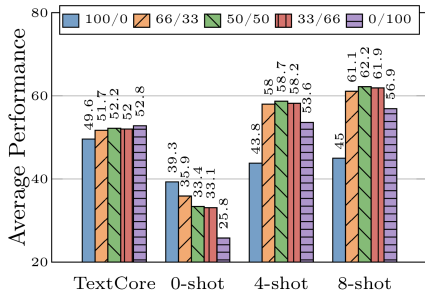
Sources: Qwen-VL, [arXiv:2308.12966v3](#), Sec. 2.1, App. E.2; InternVL 1.5, [arXiv:2404.16821v2](#), Sec. 3.1, and released code; Qwen2-VL, [arXiv:2409.12191v2](#), Sec. 2.1; Cha et al., “Honeybee: Locality-enhanced Projector for Multimodal LLM,” CVPR 2024, [arXiv:2312.06742v2](#), abstract and Fig. 2. Other rows: earlier in this lecture.

Image size, tokens	0-shot			4-shot			8-shot		
	Avg	Att	C-Abs	Avg	Att	C-Abs	Avg	Att	C-Abs
224 px, 64	30.9	30.2	30.7	55.5	52.8	55.0	58.7	56.9	62.5
336 px, 64	31.9	30.4	31.7	57.0	56.0	58.7	57.9	61.1	62.0
336 px, 144	32.2	31.5	33.4	59.1	58.2	57.4	58.7	62.3	61.2

Average over 8 captioning and VQA benchmarks. Avg / Att: average / attention pooling; C-Abs: C-Abstractor. Encoder (CLIP-trained ViT-L/14) and 1.2B LLM fixed; connector, image size and token count varied. Values from McKinzie et al., “MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training,” 2024, [arXiv:2403.09611v4](#), Figure 4; Sections 3.1–3.2 and 4.

- ▶ **Connector lesson:** “Number of visual tokens and image resolution matters most, while the type of VL connector has little effect.” Average pooling, attention pooling with learned queries and C-Abstractor (ResNet blocks with adaptive pooling, from Honeybee) “achieve very similar results at the 336px and 144 token setting” after instruction tuning.
- ▶ **Encoder lesson** (with a 2.9B LLM, Table 1): “Image resolution has the highest impact, followed by model size and training data composition”: 224 to 336 px gave about 3% on all metrics, ViT-L to ViT-H (Huge, twice the parameters) usually less than 1%. The final MM1 uses a ViT-H at 378 px and 144 tokens.
- ▶ **Cambrian-1** (Tong et al., NeurIPS 2024, [arXiv:2406.16860v2](#), Sec. 3.4) compared over 20 vision encoders: “Language supervision offers strong advantages, but the performance gap can be narrowed with SSL methods given enough data and proper tuning” (SSL: self-supervised learning, such as DINOv2).

- ▶ **Three kinds of data:** image-caption pairs (short text, closely tied to the image); interleaved image-text web documents (longer, more varied text, more loosely tied to the images, as in Flamingo); and text-only data, which preserves the LLM's language skills.
- ▶ **Data lesson 1:** "Interleaved data is instrumental for few-shot and text-only performance, while captioning data lifts zero-shot performance." Zero-shot rises from 25.8 with no captions to 39.3 with captions only; 8-shot falls from 62.2 to 45 with captions only.
- ▶ **Why:** an interleaved page already looks like a few-shot prompt, with several related images and texts in one sequence. The authors add a caveat: 3 of the 8 benchmarks are captioning, which favors caption data at zero shot.
- ▶ **Lessons 2 and 3:** text-only data lifts text-only scores, and few-shot scores when mixed with captions (with interleaved data they dip slightly); a caption/interleaved/text ratio of 5:5:1 gave strong multimodal results with comparable text-only scores.
- ▶ **Final MM1 mix:** 45% interleaved documents, 45% image-caption pairs, 10% text-only.



(a) Caption/Interleaved Mixing

Legend: caption/interleaved share of the image data (100/0: captions only). TextCore: average over 8 text-only tasks. McKinzie et al., [arXiv:2403.09611v4](#), Figure 5(a) and Section 3.3.

Counting Visual Tokens: A Worked Example

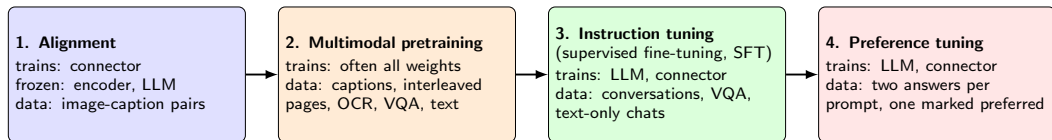
A ViT with 14-pixel patches gives $(s/14)^2$ tokens for an $s \times s$ image: 256 at 224 px, 576 at 336 px, 2,304 at 672 px.

One 1344 × 1344 photo	How the count arises	Visual tokens
BLIP-2	32 Q-Former queries at any input resolution (224 px in pretraining)	32
Flamingo	64 Perceiver Resampler latents, read through cross-attention	64
Qwen-VL	256 learned queries; the image is resized to 448 × 448	256
LLaVA-1.5	$(336/14)^2 = 24^2$; the image is resized to 336 × 336	576
LLaVA-NeXT (AnyRes)	$4 \times 576 + 576$: 2 × 2 tiles of 336 px plus the whole image at 336 px	2,880
InternVL 1.5	$(9 + 1) \times 256$: 3 × 3 tiles of 448 px plus a thumbnail (default 12-tile cap)	2,560
Qwen2-VL	$(1344/28)^2 = 48^2$: the full image at native resolution	2,304

- ▶ **Attention grows with the square.** From 576 to 2,880 visual tokens, the visual block of each head's score matrix grows from $576^2 = 331,776$ to $2,880^2 = 8,294,400$ entries (a causal mask uses about half), $(2,880/576)^2 = 25$ times. Per-token costs (FFN, stored keys and values) grow 5 times.
- ▶ **Images and frames add up.** Eight video frames at 576 tokens need 4,608 tokens, more than Vicuna v1.5's 4,096-token context; Qwen2-VL caps a training video at 16,384 tokens by lowering frame resolution.
- ▶ **Cross-attention moves the cost.** In Llama 3's image training, an image has 2,308 tokens and its text 192 on average; the image tokens pass only through the image encoder and the cross-attention layers.

Counts leave out delimiter and row-end tokens. Sources: InternVL 1.5, [arXiv:2404.16821v2](#), Sec. 3.3, and released code; Qwen2-VL, [arXiv:2409.12191v2](#), Sec. 2.1; Llama 3, [arXiv:2407.21783v3](#), Sec. 7.3. AnyRes, Vicuna: LLaVA slides.

Training an MLLM in Stages



Qwen2.5-VL

trains its new (from-scratch) ViT, LLM frozen; captions, visual knowledge, OCR; 1.5T tokens	all weights; adds interleaved, VQA, video, grounding, agent data; 2T tokens, then 0.6T at length 32,768	about 2M samples, half of them text-only; ViT frozen	image-text and text-only preference pairs, each used once; ViT frozen
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- **Curriculum:** an MLLM (multimodal LLM) meets harder data and longer inputs in later phases. Qwen2.5-VL packs 8,192-token sequences in its first two pretraining phases and 32,768-token sequences in the third.
- **Stages can merge:** InternVL3 starts from a pretrained base LLM but replaces stages 1 and 2 with one “native multimodal pre-training” stage that updates all weights on text-only and multimodal data together.

Bai et al., “Qwen2.5-VL Technical Report,” 2025, [arXiv:2502.13923v1](#), Table 2 and Sections 2.2.2, 2.3.1, 2.3.4; Zhu et al., “InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models,” 2025, [arXiv:2504.10479v3](#), Sections 2.1 and 2.2.

- **Preference tuning:** each sample x (image and question) with a preferred answer y_w and a rejected one y_l , and the model learns to prefer y_w . **DPO** (Direct Preference Optimization) learns from the pairs directly. **RLHF** (reinforcement learning from human feedback) instead fits a reward model (a network that scores answers) to the pairs and then raises that score with reinforcement learning. The DPO loss:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right)$$

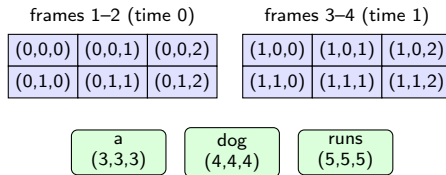
π_{θ} : the model being tuned; π_{ref} : a frozen copy of the SFT model; σ : the logistic sigmoid; β : how strongly π_{θ} is held near π_{ref} (larger β , closer). The loss is averaged over all pairs.

- **RLHF-V** builds pairs aimed at hallucination (text not grounded in the image): annotators rewrite only the wrong segments of a model's answer (1.4K answers), and a DPO variant gives the rewritten segments more weight.

y_l (model)	The clock reads approximately 11:20 ... flags flying in the top left corner ...
y_w (annotator)	The clock reads approximately 15:26 ... flags flying in the top right corner ...

Rafailov et al., "Direct Preference Optimization: Your Language Model is Secretly a Reward Model," NeurIPS 2023, [arXiv:2305.18290v3](#), Eq. 7; Yu et al., "RLHF-V: Towards Trustworthy MLLMs via Behavior Alignment from Fine-grained Correctional Human Feedback," CVPR 2024, [arXiv:2312.00849v2](#), Sections 2 and 3.1 (Eq. 5); answer excerpts from its Figure 1.

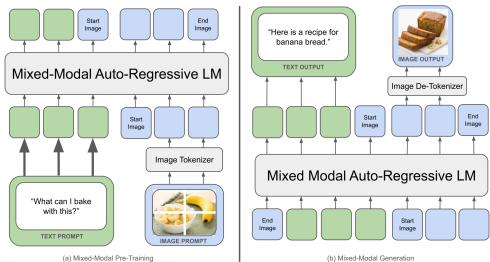
- **Native resolution:** no fixed square, no tiles. The 675M-parameter ViT reads each image at its own size and an MLP fuses each 2×2 block of 14-pixel patch tokens, so the LLM gets one token per 28×28 pixels: 64 for 224×224 (plus start and end markers), $25 \times 45 = 1,125$ for 700×1260 . LLaVA-1.5 (without HD) always uses 576.
- **RoPE** (rotary position embedding) turns each pair of query and key features by an angle proportional to the token's position, so position enters attention scores only through the relative offset. Qwen2-VL's ViT uses a 2D version over patch rows and columns.
- **Video:** sampled at 2 frames per second; a 3D convolution of depth 2 merges each pair of frames, and an image enters as two identical frames. At most 16,384 tokens per training video.



M-RoPE gives every LLM token three IDs (time, height, width), each driving its own share of the rotated feature pairs. Text gets three equal IDs, which is plain RoPE, and each new modality starts one above the largest ID so far: here a four-frame video (two merged frame pairs) of 2×3 tokens, then text. A 25×45 -token image needs 45 position values where 1D RoPE would use 1,125.

Wang et al., “Qwen2-VL: Enhancing Vision-Language Model’s Perception of the World at Any Resolution,” 2024, [arXiv:2409.12191v2](#), Section 2.1, Figure 2 (token counts); diagram adapted from Figure 3. Su et al., “RoFormer: Enhanced Transformer with Rotary Position Embedding,” 2021, [arXiv:2104.09864v5](#).

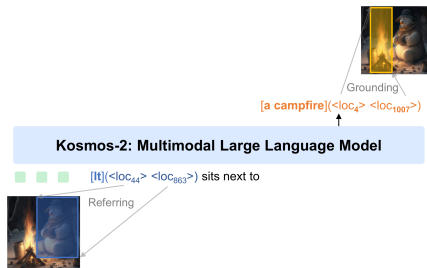
- ▶ **Early fusion:** images enter the token sequence at the first layer, with no CLIP-style encoder or connector (some texts also call LLaVA-style designs early fusion, since their visual tokens join the same sequence).
- ▶ **Fuyu-8B** (Adept, Oct. 2023): each 30×30 -pixel patch is linearly projected straight into a decoder-only Transformer, row by row, with an image-newline token after each row; any resolution works.
- ▶ **Chameleon** (Meta, 2024): a tokenizer with an 8,192-entry codebook (as in VQ-VAE, covered earlier) turns a 512×512 image into 1,024 codes in the text vocabulary, so one model reads and writes images.
- ▶ **Stability:** shared weights let norms grow until training diverged; without QK-Norm (layer norm on queries and keys), Chameleon-7B diverged after about 20% of an epoch.
- ▶ **Cost:** the tokenizer reconstructs text-heavy images poorly, which caps OCR-type tasks.



Green: text tokens; blue: image tokens. (a) Mixed-modal pretraining on a text and an image prompt; (b) mixed-modal generation of text and an image. Chameleon Team, "Chameleon: Mixed-Modal Early-Fusion Foundation Models," 2024, [arXiv:2405.09818v2](https://arxiv.org/abs/2405.09818v2), Figure 1; Sections 2.1 and 2.3.

Fuyu-8B: Adept AI, model card and config.json, huggingface.co/adept/fuyu-8b.

- ▶ **Kosmos-2** (Peng et al., 2023) links phrases to boxes (grounding) with one new token per cell of a 32×32 grid, 1,024 in all. A box is its top-left and bottom-right cells, placed after its phrase like a Markdown link. Cell = $32 \cdot \text{row} + \text{column}$, counting from 0, so `<loc44>` is row 1, column 12.
- ▶ **Qwen-VL** adds no location tokens: coordinates are scaled to $[0, 1000)$ and written as plain text, as in `<ref>the ear on a giraffe</ref>` `<box>(176,106) , (232,160)</box>`.
- ▶ **Qwen2.5-VL** uses the image's actual pixel coordinates for boxes and points, so the numbers also carry object size.
- ▶ **Scoring** referring expressions (RefCOCO and its variants): a box counts as correct if its IoU (covered earlier) with the ground-truth box exceeds 0.5.



Peng et al., "Kosmos-2: Grounding Multimodal Large Language Models to the World," 2023, [arXiv:2306.14824v3](https://arxiv.org/abs/2306.14824v3), Figure 1; Sections 3.1, 3.3 and 4.1.2.

Cell order: Kosmos-2 decoding script, github.com/microsoft/unilm. Bai et al., "Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond," 2023, [arXiv:2308.12966v3](https://arxiv.org/abs/2308.12966v3), Section 2.2, Appendix B.1; Qwen2.5-VL: [arXiv:2502.13923v1](https://arxiv.org/abs/2502.13923v1), Sections 2.1.2, 2.2.1.

Open-Weight MLLMs, 2024–2025: One Blueprint

Model	Vision encoder	Connector	LLM	Visual tokens per image
LLaVA-OneVision	SigLIP SO400M (384 px)	2-layer MLP	Qwen2 0.5B, 7B, 72B	AnyRes: 384-px crops (grids up to 6×6) plus the full view, 729 tokens each, interpolated down to $\leq 7,290$ in all
Qwen2.5-VL	own ViT: window attention in most layers, 2D-RoPE	2×2 merge, 2-layer MLP	Qwen2.5 3B, 7B, 72B	native size: one per 28×28 pixels
InternVL3	InternViT-300M or -6B (448 px)	pixel unshuffle (2×2 into 1; InternVL 1.5 calls it pixel shuffle), 2-layer MLP	Qwen2.5 (0.5B–72B) or InternLM3-8B	256 per 448-px tile; tile grid fitted to the aspect ratio (by default up to 12) plus a thumbnail
Gemma 3	SigLIP 400M (896 px), frozen	4×4 average pooling, linear map	Gemma 3 4B, 12B, 27B	256 per 896-px view; at inference, Pan & Scan splits large or non-square images into crops

Every row keeps LLaVA's layout. Beyond the choice of parts, the rows differ most in the visual-token budget and the last training step: instruction tuning for LLaVA-OneVision, DPO for Qwen2.5-VL, mixed preference optimization (DPO plus two more losses) for InternVL3, and distillation plus reinforcement learning for Gemma 3.

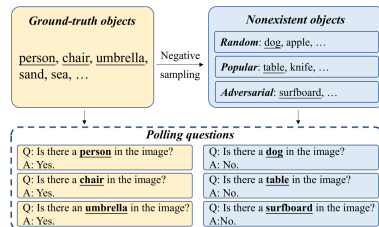
Li et al., [arXiv:2408.03326v3](#); Bai et al., [arXiv:2502.13923v1](#); Zhu et al., [arXiv:2504.10479v3](#) and the [InternVL3-8B model card](#); Gemma Team, [arXiv:2503.19786v1](#), and [Transformers v4.50.0](#) (linear map).

Benchmark	What it tests	Format	Metric
VQAv2 CVPR 2017	Most questions get a similar photo with a different answer	about 1.1M image-question pairs on COCO	VQA accuracy
TextVQA CVPR 2019	Reading text in photos: signs, brands, times	45,336 questions on 28,408 images	VQA accuracy
DocVQA WACV 2021	Reading document pages, where forms and tables carry meaning	50,000 questions on 12,767 page images	ANLS: average normalized Levenshtein similarity
MMMU CVPR 2024	College-level questions in 30 subjects and 30 image types	11,550 questions, 94% multiple choice	accuracy pooled over all questions (micro-average)
MMBench ECCV 2024	20 perception and reasoning abilities, in English and Chinese	3,217 multiple-choice questions	CircularEval: right under every option rotation
POPE EMNLP 2023	Object hallucination: naming objects that are not there	yes/no: "Is there a ... in the image?"	accuracy, F1, share of "yes"
MMVP CVPR 2024	Details that CLIP encoders miss, such as orientation and counts	300 two-option questions on 150 image pairs	pair accuracy: both questions right

- **ANLS** gives partial credit for near-miss strings. **CircularEval** repeats each question once per rotation of its options and needs every answer right: OpenFlamingo v2 drops from 36.7% to 2.6% on MMBench-dev.

Goyal et al., [arXiv:1612.00837v3](#); Singh et al., [arXiv:1904.08920v2](#); Mathew et al., [arXiv:2007.00398v3](#); Yue et al., [arXiv:2311.16502v4](#); Liu et al., [arXiv:2307.06281v5](#), Table 2; Li et al., [arXiv:2305.10355v3](#); Tong et al., [arXiv:2401.06209v2](#).

- **Object hallucination:** naming objects absent from the image. POPE (Polling-based Object Probing Evaluation) asks “Is there a ⟨object⟩ in the image?”, half about annotated objects (yes), half about absent ones (no).
- **Absent objects** are drawn at random (random), from the dataset’s most frequent objects (popular), or from those that co-occur most with the objects present (adversarial).
- **Metrics** (“yes” is positive): accuracy, precision, recall, F1, share of “yes”. Always saying yes gives 50% accuracy and $F1 = 66.67$.
- **It follows the instruction data** (mostly built on COCO): about half of hallucinated objects are among COCO’s ten most frequent, over half among the ten that co-occur most with objects present. InstructBLIP, tuned on short public-dataset instructions, hallucinates least.



Li et al., “Evaluating Object Hallucination in Large Vision-Language Models,” EMNLP 2023, [arXiv:2305.10355v3](https://arxiv.org/abs/2305.10355v3), Figure 3 (right part); Sections 3.2 and 4.3.

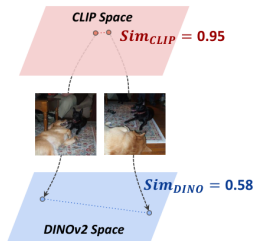
		Random	Popular	Advers.
LLaVA	accuracy	54.43	52.43	50.77
	“yes” (%)	95.37	97.37	99.10
InstructBLIP	accuracy	88.73	81.37	74.37
	“yes” (%)	55.20	62.57	68.97

COCO val, 500 images, 6 probes each (Table 3).

Step 1

Finding CLIP-blind pairs.

Discover image pairs that are proximate in CLIP feature space but distant in DINOv2 feature space.



Step 2

Spotting the difference between two images.

For a CLIP-blind pair, a human annotator attempts to spot the visual differences and formulates questions.

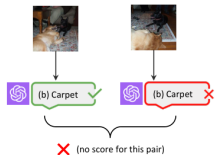


Step 3

Benchmarking multimodal LLMs.

Evaluate multimodal LLMs using a CLIP-blind image pair and its associated question.

Where is the yellow animal's head lying in this image?
(a) Floor (b) Carpet



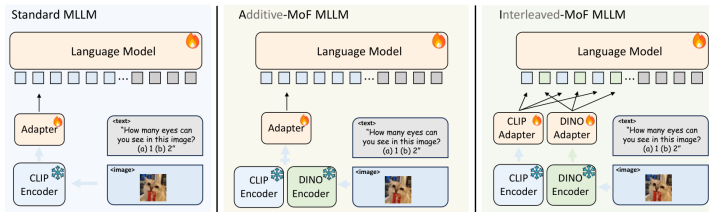
The model receives a score only when **both** predictions for the CLIP-blind pair are correct.

Tong et al., "Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs," CVPR 2024, [arXiv:2401.06209v2](https://arxiv.org/abs/2401.06209v2), Figure 2; thresholds from Section 2.1.

- **CLIP-blind pair:** two images with cosine similarity above 0.95 in CLIP ViT-L/14 but below 0.6 in DINOv2 ViT-L/14, a vision-only model (covered earlier) that keeps more detail.

- ▶ **MMVP:** annotators wrote 300 two-option questions about the visible difference in 150 CLIP-blind pairs. A pair scores only if both answers are right, so guessing gives 25%.
- ▶ **Results:** humans 95.7%; Gemini 40.7%, GPT-4V (GPT-4 with image input) 38.7%; LLaVA-1.5 24.7%, InstructBLIP 16.7%, LLaVA (336 px) 6.0%. Every other model tested is below chance.
- ▶ **Nine failure patterns**, among them orientation, counting and viewpoint: CLIP features favor high-level semantics over detail, the image-side twin of the bag-of-words behavior seen earlier in this lecture.

Tong et al., [arXiv:2401.06209v2](#), Sections 2.2, 2.3 and 3.1, Figure 4.



Snowflake: frozen; flame: trained. Tong et al., [arXiv:2401.06209v2](https://arxiv.org/abs/2401.06209v2), Figure 7; Section 3.2, Tables 1–3 and 5.

- **Scale does not fix it:** on MMVP-VLM, where the questions become captions that CLIP must match to the right image, ten CLIP-style encoders (OpenAI CLIP, SigLIP, MetaCLIP, EVA-CLIP, and DFN, a CLIP trained on data chosen by a data filtering network) gain from more data and parameters on only 2 of the 9 patterns.
- **The MLLM inherits the errors:** across the nine patterns, the accuracies of CLIP and of LLaVA-1.5 correlate with Pearson $r = 0.87$.
- **Additive MoF (224 px):** one adapter reads $\alpha \cdot \text{CLIP} + (1 - \alpha) \cdot \text{DINOv2}$. At $1 - \alpha = 0.75$, MMVP rises from 5.5 to 18.7 but LLaVA-Bench drops from 81.8 to 75.8.
- **Interleaved MoF:** one adapter per encoder, the two token streams alternating in spatial order (224 px, 512 tokens). Against the 336-px, 576-token models: MMVP 6.0 to 16.7 (LLaVA) and 24.7 to 28.0 (LLaVA-1.5); LLaVA-Bench 81.4 to 82.8 and 84.7 to 82.7.

- ▶ **An old problem:** on the original VQA dataset (v1), answering “yes” to every “Do you see a...” question scored 87% VQA accuracy. On balanced VQAv2 a question-only model still scores 43.01% (best model with the image: 59.14%).
- ▶ **The image is often unnecessary:** all 22 LLMs tested answer “What is the capital of Nebraska?” (asked over a map) blind. In ScienceQA, 57.2% of the questions are answered blind by six or more of eight LLMs.
- ▶ **Leakage:** an MLLM that answers blind well above its own LLM base has probably seen the items in multimodal training (table).
- ▶ **Rule:** always report blind baselines, the model with the image removed (or a gray image) and its LLM alone.

- ▶ **MMStar** keeps only samples that at most two of eight LLMs answer without the image (22,401 down to 11,607); three experts then pick 1,500 that need the image, 250 for each of six capabilities. LLMs score near random on it.

MMM validation, %	GPT-4V	Sphinx-X-MoE
LLM base alone, S_t	41.2	25.7
image removed, S_{ov}	45.1	43.6
with the image, S_{wv}	53.6	44.8
$MG = S_{wv} - S_{ov}$	8.5	1.2
$ML = \max(0, S_{ov} - S_t)$	3.9	17.9

MG, multimodal gain: what the image adds. ML, multimodal leakage: how far the blind model beats its LLM base (GPT-4 Turbo, Mixtral-8x7B). Random choice: 22.1.

Goyal et al., [arXiv:1612.00837v3](https://arxiv.org/abs/1612.00837v3), Section 1, Table 1; Chen et al., “Are We on the Right Way for Evaluating Large Vision-Language Models?,” NeurIPS 2024, proceedings.neurips.cc, Sections 3 and 4, Figures 1, 2, 4, Tables 2, 4, Appendices A.1 and A.3.

- ▶ **Hallucination:** multimodal LLMs still describe objects and details that are not in the image, most often objects that are frequent in their training data or that often co-occur there with the objects actually present. Better data and preference tuning reduce the rate but do not eliminate it.
- ▶ **A perception bottleneck:** most systems see the image through one contrastively trained encoder at a fixed patch size. Whatever that encoder conflates (orientation, counts, small text, spatial relations), the language model cannot recover.
- ▶ **Composition:** contrastive training rewards matching the right words more than the right structure, so relations, attributes and word order remain weak points of the joint embedding space.
- ▶ **Cost:** with per-patch connectors, visual tokens grow with resolution, the number of images and the number of video frames (resamplers fix the count but compress detail), and attention cost grows with the square of the sequence length. Every resolution strategy is a trade between detail and context budget.
- ▶ **Evaluation:** benchmark questions leak into training data, many can be answered from language priors alone, and multiple-choice formats overstate perception; a blind (text-only) baseline is the minimum control.
- ▶ **Data:** web image-text pairs carry the biases and errors of their sources, and generated instruction data inherits the mistakes of the model that wrote it.

- ▶ **A vision-language model returns a score or text.** Dual encoders (CLIP, ALIGN, SigLIP) score and retrieve; fusion models (ALBEF, BLIP, CoCa) also match pairs, and BLIP and CoCa caption; LLM-based models (Flamingo, BLIP-2, LLaVA, Qwen-VL) write open-ended text.
- ▶ **The loss shapes the joint space.** CLIP normalizes over the batch; SigLIP scores each pair with a sigmoid and wins clearly at small batches. The space keeps a modality gap, reads captions almost as bags of words, and gains more from curated data than from more data (same compute: LAION-2B's 2.3B pairs 73.1%, DataComp-1B's 1.4B filtered pairs 79.2%).
- ▶ **BLIP trains one network three ways** (ITC, ITM with hard negatives, LM) and cleans its own data with CapFit; BLIP-2 trains only a 188M-parameter Q-Former (plus one linear layer) whose 32 query outputs prompt a frozen LLM.
- ▶ **Flamingo adds vision to a frozen LM:** 64 resampled visual tokens per image, tanh-gated cross-attention that starts as the unchanged LM, and per-image attention masking over interleaved web pages for few-shot prompting.
- ▶ **LLaVA made the recipe simple:** 158K instructions written by a text-only GPT-4, a projection into Vicuna, two training stages; LLaVA-1.5 (MLP, 336 px, format prompts, 1.2M public samples) and image tiling followed.
- ▶ **Extending an LLM with vision is four decisions** (encoder, connector, entry point, token budget). In MM1, resolution and token count matter most and the connector type has little effect; attention cost grows with the square of the token count.
- ▶ **MLLMs train in stages:** alignment, multimodal pretraining, instruction tuning, preference tuning (DPO). Native resolution with M-RoPE and boxes written as text tokens extend the same blueprint; early fusion drops the vision encoder and feeds patches or image codes straight to the LLM.
- ▶ **Evaluation must separate perception from language priors:** match the benchmark to the skill, probe hallucination (POPE) and CLIP-blind pairs (MMVP: humans 95.7%, GPT-4V 38.7%), and report blind baselines.

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